Faculty of Computer Applications and IT

Subject: Embedded Systems (230701302)

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Chapter – 1 : A Systems Engineering Approach to Embedded Systems Design

Delivered by: Dr. Krupa Mehta

- This material is prepared for the students of MCA, FCAIT – PG Programme, GLS University considering the syllabus for the subject “Embedded Systems”
- The presentation included here outlines only the main points and key highlights of the course. It does not cover the entire syllabus in detail. For comprehensive information and complete understanding of the course material, please refer to the full syllabus and associated readings.

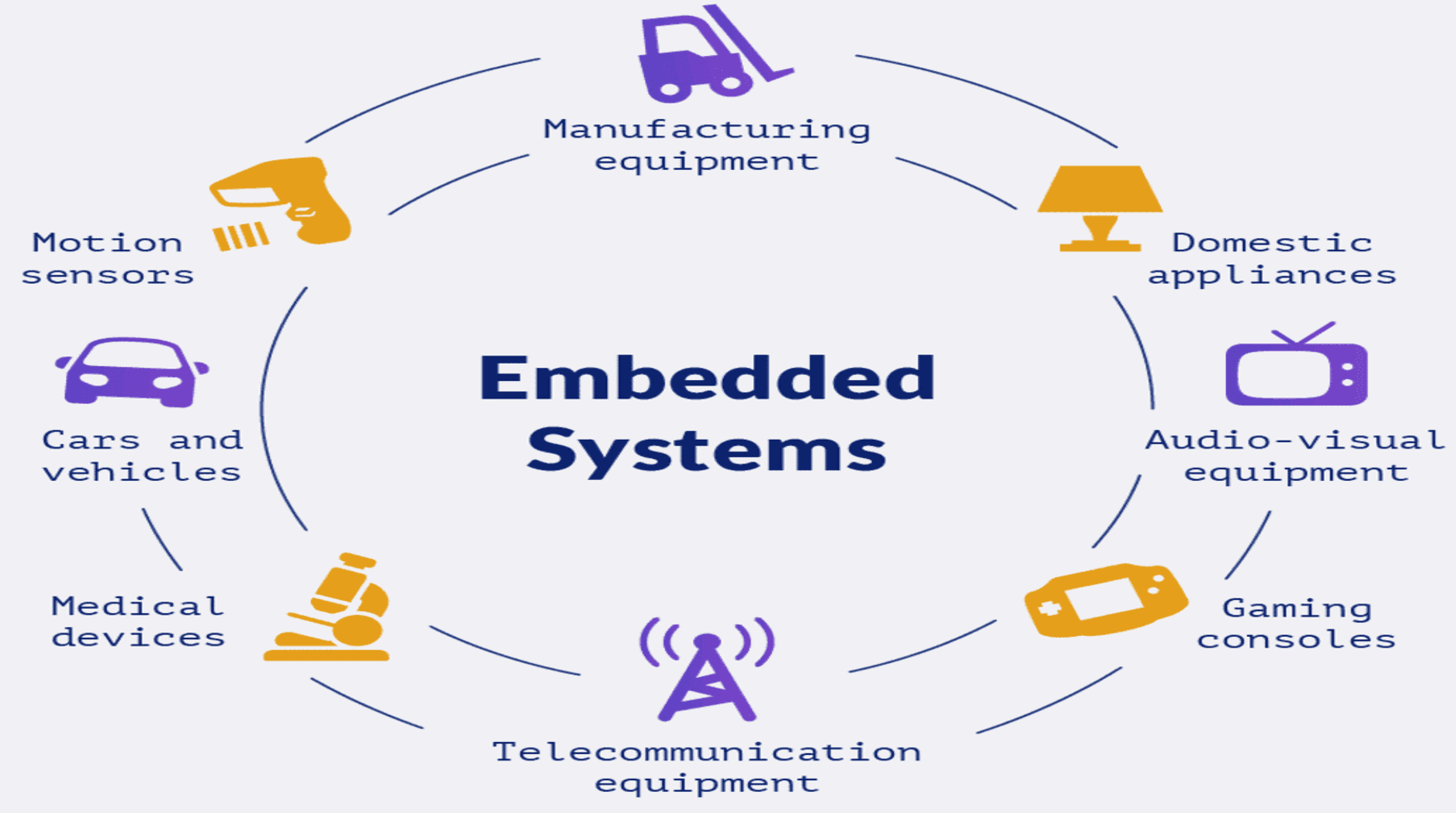
Embedded Systems

- An embedded system is an applied computer system distinct from PCs or supercomputers
  - Limited Hardware/Software
  - Dedicated Function
  - High Quality/Reliability

Embedded Systems

- Embedded systems are more limited in hardware and/or software functionality than a personal computer
- An embedded system is designed to perform a dedicated function
- An embedded system is a computer system with higher quality and reliability requirements than other types of computer systems
- Some devices that are called embedded systems, such as PDAs or web pads, are not really embedded systems

Embedded Systems

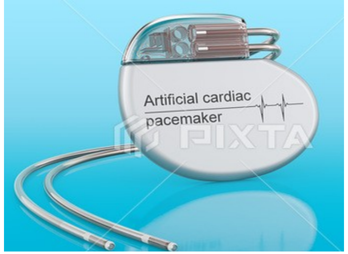


Automotive Embedded Systems

Consumer Electronics Embedded Systems

Industrial Control Embedded Systems

Medical Embedded Systems



Pacemaker

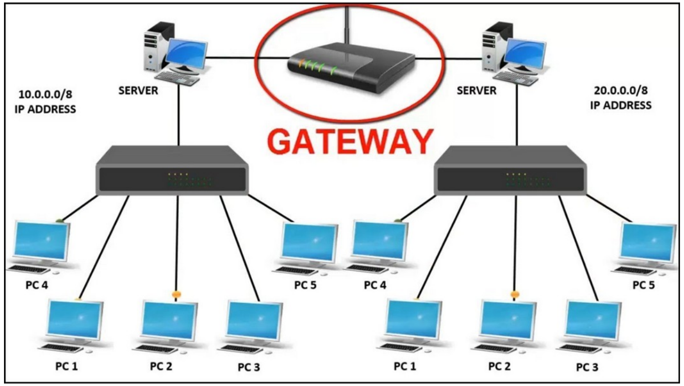
Infusion Pump

Dialysis Machines

Prosthetic Devices

Cardiac Monitors

Networking Embedded Systems

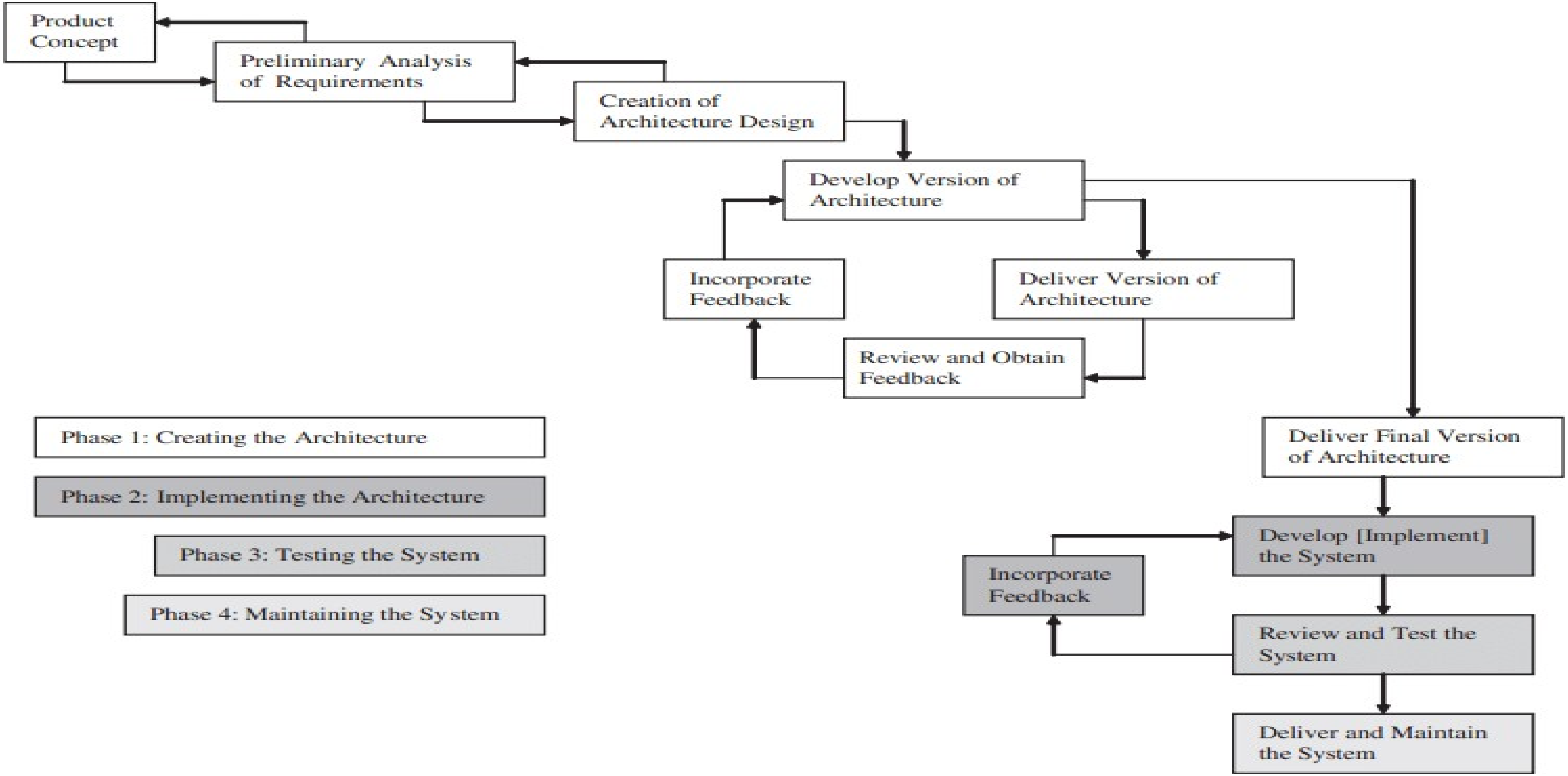


Office Automation Embedded Systems

Embedded Systems Design

- The big-bang model
  - no planning or processes in place before and during the development of a system
- The code-and-fix model
  - product requirements are defined but no formal processes are in place before the start of development
- The waterfall model
  - a process for developing a system in steps, where results of one step flow into the next step
- The spiral model
  - process for developing a system in steps, and throughout the various steps, feedback is obtained and incorporated back into the process

Embedded Systems Design and Development Lifecycle Model



Embedded Systems Architecture

- Structure Types
  - Uses (also referred to as subsystem and component)
    - Layers
    - Kernel
    - Channel Architecture
    - Virtual Machine
  - Decomposition
  - Class (also referred to as generalization)

Embedded Systems Architecture

- Component and Connector
  - Client/Server (also referred to as distribution)
  - Process (also referred to as communicating processes)
  - Concurrency and Resource
    - Interrupt
    - Scheduling (EDF, priority, roundrobin)
  - Memory
    - Garbage Collection
    - Allocation
  - Safety and Reliability

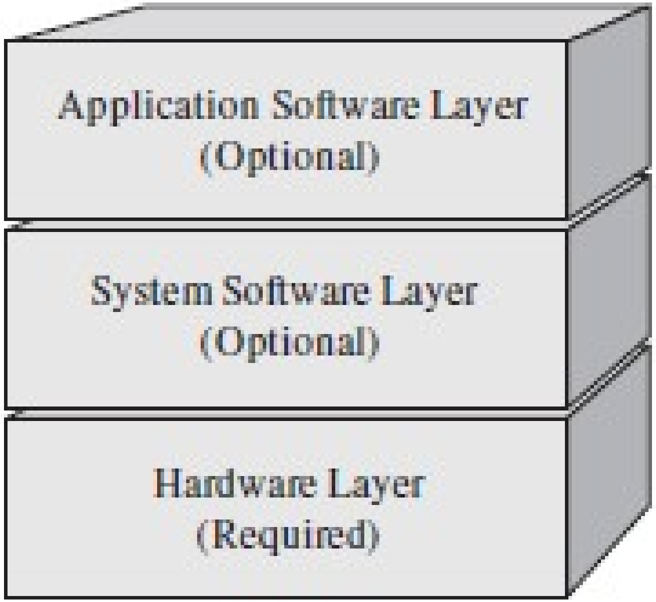
Embedded Systems Architecture

- Allocation
  - Work Assignment
  - Implementation
  - Deployment

Importance of Embedded System Architecture

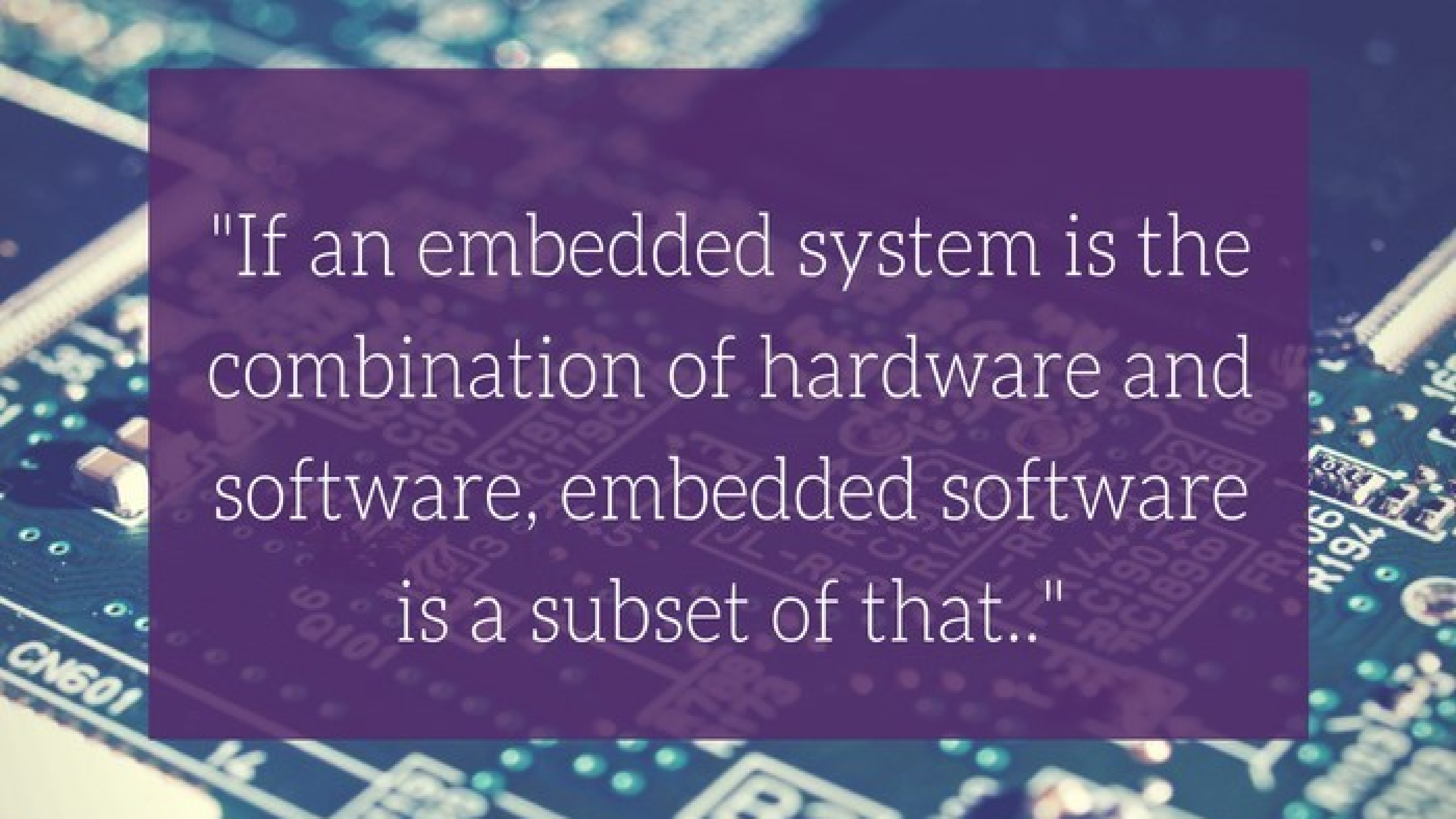
- defining and capturing the design of a system
- cost limitations
- determining a system’s integrity, such as reliability and safety
- working within the confines of available elemental functionality (i.e., processing power, memory, battery life, etc.)
- marketability and sellability
- deterministic requirements
- every embedded system has an architecture
- an embedded architecture captures various views

The Embedded Systems Model



Reference

- Tammy Noergaard (2005). Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers. Elsevier


"If an embedded system is the combination of hardware and software, embedded software is a subset of that.."